Wonjun Jo

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Website | Scholar | GitHub | LinkedIn

Research Interests

Learning and adaptation for embodied agents through self-supervised and multimodal learning. I study adaptation of **pretrained robot policies** and acquisition of **reusable skills** from demonstrations and interaction, with interests in learning from failures, preserving prior capabilities, and generalizing to new tasks and environments.

Current Research

Wonjun Jo, Nam Hyeon-Woo, Yohan Park, Hyunwoo Ha, Tae-Hyun Oh, "When Fine-Tuning Drifts: Mitigating Policy Drift with Base Action-Field Regularization," ***Under review***, 2026.

- Mitigating policy drift for robust few-shot fine-tuning of pretrained robot policies.

Wonjun Jo, "One-shot Teaching for Reusable Agentic Robot Skills," ***In progress***.

Education

POSTECH — Pohang, South Korea
Ph.D. Candidate, Electrical Engineering (MS-Ph.D. Integrated; Advisor: Tae-Hyun Oh) — *Mar. 2022 – Present*

Inha University — Incheon, South Korea
B.S. in Computer Engineering — *Mar. 2018 – Feb. 2022*

Peer-Reviewed Publications

[1] **Wonjun Jo**, Hyunwoo Ha, Kim Ji-Yeon, Hawook Jeong, Tae-Hyun Oh, "LiDAR-Anchored Collaborative Distillation for Robust 2D Representations," *IEEE Robotics and Automation Letters (**RA-L**)*, 2026. [Project Page] [Paper] [arXiv]

- Improving the robustness and generalization of self-supervised visual representations with LiDAR guidance.

[2] Yohan Park, Hyunwoo Ha, **Wonjun Jo**, Tae-Hyun Oh, "DarkQA: Benchmarking Vision-Language Models on Visual-Primitive Question Answering in Low-Light Indoor Scenes," *IEEE Robotics and Automation Letters (**RA-L**)*, 2026. [Project Page] [Paper] [arXiv]

- Benchmarking the robustness of embodied vision-language models under low-light conditions.

[3] **Wonjun Jo**, Hyunwoo Ha, Kim Ji-Yeon, Hawook Jeong, Tae-Hyun Oh, "Self-Supervised Collaborative Distillation: Enhancing Lighting Robustness and 3D Awareness," Workshop on Wild3D, *IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2025. [Paper] [OpenReview]

- Improving pretrained visual representations for lighting robustness and 3D-aware generalization.

[4] **Wonjun Jo**, Kwon Byung-Ki, Kim Ji-Yeon, Hawook Jeong, Kyungdon Joo, Tae-Hyun Oh, "The Devil is in the Details: Simple Remedies for Image-to-LiDAR Representation Learning," *Asian Conference on Computer Vision (**ACCV**)*, 2024. [Project Page] [Paper] [arXiv]

- Improving self-supervised image-to-LiDAR representation learning for robust 3D perception.

Research Experience

KIST Physical AI Research Intern — Seoul, South Korea
Korea Institute of Science and Technology (Advisor: In So Kweon) — *Jul. 2026 – Present*

POSTECH → KAIST Research Assistant — Pohang → Daejeon, South Korea
AMI Lab, POSTECH → KAIST (lab moved in 2025; Advisor: Tae-Hyun Oh) — *Feb. 2022 – Present*

POSTECH Undergraduate Research Intern — Pohang, South Korea
Pohang University of Science and Technology (Advisor: Tae-Hyun Oh) — *Jun. 2021 – Aug. 2021*

Selected Research Projects

Modular and Agentic Physical AI (KIST) Jul. 2026 – Present

- Working on modular robot architectures that link perception, planning, and skill execution through agentic programming.
- Studying robust vision and tactile feedback for dexterous manipulation with high-DoF hands.
- Using cooking as a testbed for long-horizon planning, task-state tracking, and precise bimanual manipulation.

Self-Supervised Learning Method Utilizing Camera-LiDAR Correlation for Autonomous Driving Jan. 2023 – Dec. 2024

- Proposed a novel self-supervised learning method for improving 2D representations using LiDAR.
- Proposed simple remedies for improving self-supervised image-to-LiDAR distillation.
- Funded by RideFlux Inc., South Korea.

Deformable Object Recognition Technology Research Mar. 2022 – Dec. 2022

- Developed a deformable object recognition algorithm for multi-spectral images.
- Developed a motion vector estimation algorithm for deformable objects using LiDAR.
- Funded by the Agency for Defense Development (ADD), South Korea.

Research Infrastructure & Robot Setup

KIST 2026 – Present

- Purchased and set up a Franka Research 3 arm with a Robotiq 2F-85 gripper, and a Bi-Nero dual-arm robot with Revo2 Touch hands.
- Currently run real-robot experiments on the Franka Research 3 setup.

POSTECH → KAIST 2022 – Present

- Purchased and set up GPU servers and storage/NAS systems at POSTECH and KAIST.
- Purchased a Franka Research 3 arm and a Mobile ALOHA.

Patents

Self-Supervised Learning Method, Computing Device, and Computer Program Through Collaborative Distillation Between 2D and 3D Encoders 2024

- Republic of Korea. Application No. 10-2024-0202291; Patent No. 10-2848095 (2025).

Self-Supervised Learning Method, Computing Device, and Computer Program Using Synchronized Sensor Data and Unsynchronized Sensor Data 2023

- Republic of Korea. Application No. 10-2023-0146900; Patent No. 10-2857995 (2025).

Academic Service

Journal Reviewer

- The Visual Computer: International Journal of Computer Graphics (**TVCJ**), 2023

Conference Reviewer

- International Conference on Robotics and Automation (**ICRA**), 2023

Teaching Experience

[EECE454] Introduction to Machine Learning, POSTECH, 2022